

PULIN[®] FILM COATED TABLET 10MG

Ingredient(s):
Each tablet contains:
Metoclopramide HCl 10mg

- Pharmacodynamics:**
1. Metoclopramide has at least two pharmacological actions at clinical doses. It is an antagonist at D2 dopamine receptors in the chemoreceptor trigger zone. It also has actions on muscarinic cholinergic system within the gut increasing the release of acetylcholine.
 2. It inhibits gastric smooth muscle relaxation produced by dopamine thus enhancing cholinergic responses of GI smooth muscle.
 3. It accelerates intestinal transit and gastric emptying by preventing relaxation of gastric body and increasing the phasic activity of antrum. At the same time, this action is accompanied by relaxation of the upper small intestine, resulting in an improved coordination between the body and antrum of the stomach and the upper small intestine.
 4. Metoclopramide decreases reflux into the esophagus by increasing the resting pressure of the lower esophageal sphincter and improves acid clearance from the esophagus by increasing amplitude of esophageal peristaltic contraction.
 5. The dopamine antagonist action raises the threshold of activity in the vomiting center and decreases the input from afferent visceral nerves.

- Pharmacokinetics:**
1. Metoclopramide is almost completely absorbed following oral dosing, and peak plasma concentrations are seen within 0.5 to 2 hours in fasted healthy volunteers. Metoclopramide undergoes first-pass metabolism with bioavailability of between 32 to 98%.
 2. Protein binding of metoclopramide is around 40%. The clearance is principally by hepatic metabolism; only 20% is excreted in urine unchanged. The half-life of metoclopramide after oral administration is 4.5 hours.

- Indication(s):**
Adult population
This product is indicated for use in adults for:
- Prevention of delayed chemotherapy induced nausea and vomiting (CINV)
 - Prevention of radiotherapy induced nausea and vomiting (RINV)
 - Symptomatic treatment of nausea and vomiting including acute migraine induced nausea and vomiting

- Pediatric population
This product is indicated in children (aged 1 to 18 years) for:
- Prevention of delayed chemotherapy induced nausea and vomiting (CINV) as a second-line option

Dosage and Administration:
Adults
The recommended single dose is 10mg, repeated up to three times daily.

The maximum recommended daily dose is 30mg or 0.5mg/kg body weight.
The maximum recommended treatment duration is 5 days.

Prevention of delayed chemotherapy induced nausea and vomiting (CINV)(pediatric patients aged 1-18 years)
The recommended dosage is 0.1 to 0.15mg/kg body weight, repeated up to 3 times daily by oral route. Total daily dose of metoclopramide, especially for children and young adults, should not normally exceed 0.5mg/kg body weight.

Dosing table

Age	Body Weight	Dose	Frequency
1 - 3 years	10 - 14kg	1mg	Up to 3 times daily
3 - 5 years	15 - 19kg	2mg	Up to 3 times daily
5 - 9 years	20 - 29kg	2.5mg	Up to 3 times daily
9 - 18 years	30 - 60kg	5mg	Up to 3 times daily
15 - 18 years	Over 60kg	10mg	Up to 3 times daily

The maximum treatment duration is 5 days for prevention of delayed chemotherapy induced nausea and vomiting (CINV).

Tablets are not suitable for use in children weighing less than 30kg. Other pharmaceutical forms may be more appropriate for administration for this population.

Frequency of administration.
A minimum interval of 6 hours between two administrations is to be respected, even if vomiting or rejection of the dose occurs.

Special populations
Elderly subjects
In elderly patients a dose reduction should be considered, based on renal and hepatic function, and overall frailty.

Renal impairment
In patients with end stage renal disease (Creatinine clearance ≤ 15 ml/min), the dose should be reduced by 75%.
In patients with moderate to severe renal impairment (Creatinine clearance between 15-60ml/min), the dose should be reduced by 50%.

Hepatic Impairment
In patients with severe hepatic impairment, the dose should be reduced by 50%.

Route of Administration:
For oral administration.

- Contraindication(s):**
- Hypersensitivity to the active substance or to any of the excipients listed
 - Gastrointestinal haemorrhage, mechanical obstruction or gastro-intestinal perforation for which the stimulation of gastrointestinal motility constitutes a risk
 - Confirmed or suspected pheochromocytoma, due to the risk of severe hypertension episode.

- History of neuroleptic or metoclopramide-induced tardive dyskinesia
- Epilepsy (increase crises frequency and intensity)
- Parkinson's disease
- Combination with levodopa or dopaminergic agonist
- Known history of methaemoglobinemia with metoclopramide or of NADH cytochrome-b5 deficiency
- Use in children less than 1 year of age

Precaution(s) / Warning(s):

Avoid doses exceeding 0.5mg/kg/day

Neurological disorder

- Extrapyramidal effects, especially dystonic reaction of metoclopramide are more likely to occur in children shortly after initiation of therapy, and usually with doses higher than 0.5mg per kg of body weight per day. These reactions generally occur at the beginning of treatment, and can occur after a single dose. If extrapyramidal symptoms occur, metoclopramide should be discontinued immediately. These effects are generally completely reversible after treatment discontinuation, however, symptomatic treatment may be required (benzodiazepines in children, and/or anticholinergic antiparkinsonian medicinal products in adults).
- An interval of at least six hours should be respected between each dose even if vomiting or rejection of the dose occurs, in order to avoid overdose.
- Long-term treatment with metoclopramide may cause potentially irreversible tardive dyskinesia, particularly in elderly subjects. Treatment should not exceed 3 months because of the risk of tardive dyskinesia. Treatment must be discontinued if clinical signs of tardive dyskinesia occur.
- Neuroleptic malignant syndrome has been described with metoclopramide in combination with neuroleptics and with metoclopramide monotherapy. Metoclopramide must be immediately discontinued if symptoms of neuroleptic malignant syndrome develop, and appropriate treatment should be initiated.
- Particular caution should be exercised in patients with underlying neurological disorders, and in patients receiving other centrally-acting drugs.
- Symptoms of Parkinson's disease may also exacerbated by metoclopramide.

Methemoglobinemia

- Methemoglobinemia, which could be related to NADH-cytochrome b5 reductase deficiency, has been reported. If this occurs, treatment must be immediately and permanently discontinued, and appropriate measures initiated (such as treatment with methylene blue)

Cardiac disorders

- Serious cardiovascular undesirable effects, including cases of severe bradycardia, circulatory collapse, cardiac arrest and QT prolongation have been reported during administration of metoclopramide by injection, particularly via the intravenous route.
- Particular caution should be exercised when administering metoclopramide, particularly via the intravenous route, in elderly subjects, patients with cardiac conduction disorders (including QT prolongation), patients with electrolyte imbalance, bradycardia and patients taking other drugs known to prolong QT interval.
- The intravenous injection must be given as slow bolus (of at least 3 minutes' duration) in order to reduce the risk of undesirable effects (e.g. hypotension, akathisia)

Kidney or liver failure

- In patients with kidney failure or severe liver failure, a dose reduction is recommended.

Drug Interaction(s):

1. Concurrent use of anticholinergics and opioid-containing medication may antagonize the effects of metoclopramide on gastrointestinal motility.
2. Extrapyramidal reaction - causing medication such as phenothiazines & haloperidol if used concurrently with metoclopramide, may increase the frequency & severity of extrapyramidal effects such as akathisia and dystonia.
3. Concurrent use with alcohol and CNS depressant may increase the sedative effects of either the CNS depressant or metoclopramide.

Pregnancy and lactation:

It is not recommended for use in pregnancy unless there are compelling reasons. It is not advised for use during the first trimester.

Side Effect(s) / Adverse Reaction(s):

The most common adverse effect and due to the action of dopamine antagonist, including, drowsiness, fatigue, dizziness, hyperprolactinaemia, and extrapyramidal reaction such as akathisia and dystonia may occasionally occur.

Symptoms and Treatment for Overdosage and Antidote(s):

Symptoms of overdosage: may include severe drowsiness, confusion, disorientation and extrapyramidal reactions such as dystonic reaction.

The recommended treatment of overdosage includes:

Anticholinergic or antiparkinson drugs or antihistamine with anticholinergic properties (50mg of diphenhydramine administered intramuscularly in adults and 1mg per kg of body weight [mg/kg] intramuscularly or intravenously in infants and children) to help in controlling the extrapyramidal reactions.

Effects on Ability to Drive and Use Machine:

Metoclopramide may cause drowsiness, dizziness, dyskinesia and dystonias which could affect the vision and also interfere with the ability to drive and operate machinery.

Shelf-Life:

3 years from the date of manufacture.

Storage Condition(s):

Store at temperature below 30°C. Protect from light and moisture.

Product Description & Packing(s):

A pale orange film coated tablet.

Blister packing of 10's x 10 and 10's x 100.



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